Deliverables: Energy-Efficient Photonic Neurons via High-Q Microring Resonators

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Figure 1

SPM phase shift in straight Si_3N_4 waveguide (Agrawal Eqs. 4.1.5–4.1.7)
$\gamma \approx 1\,W^{-1}m^{-1}$ → **π-shift requires watts at cm scale, kW at mm scale**

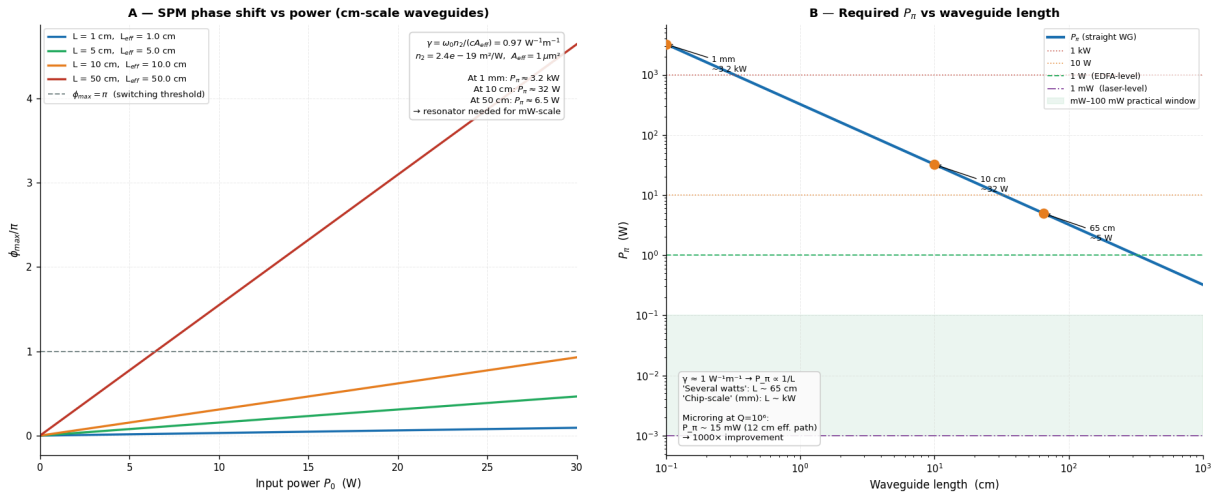


Figure 2

Microring resonance lineshape — Bogaerts all-pass transfer function
$T(\delta) = (a^2 - 2ar\cos\delta + r^2)/(1 - 2ar\cos\delta + (ar)^2)$

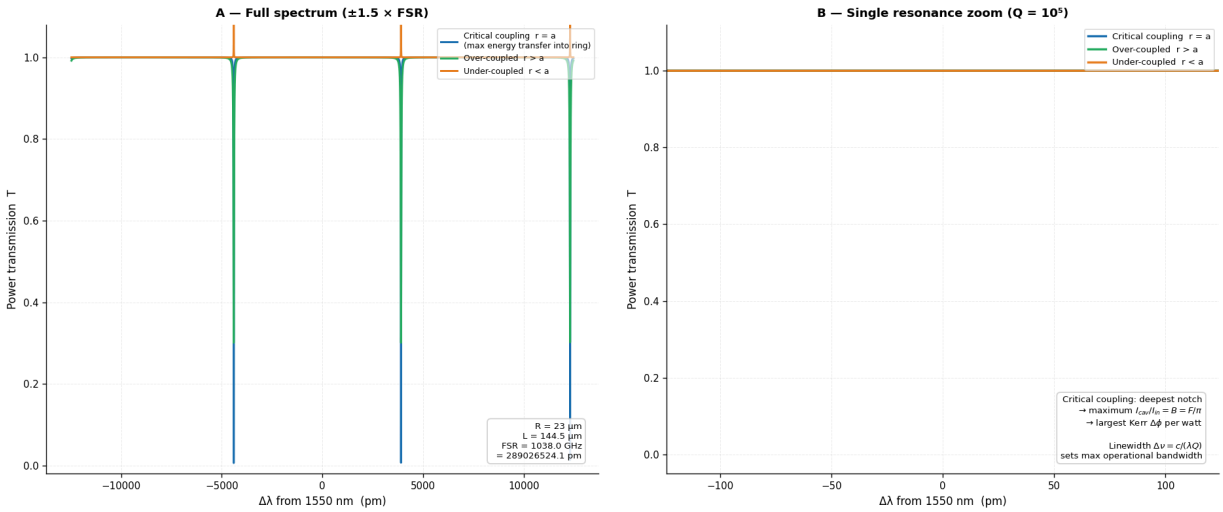


Figure 3

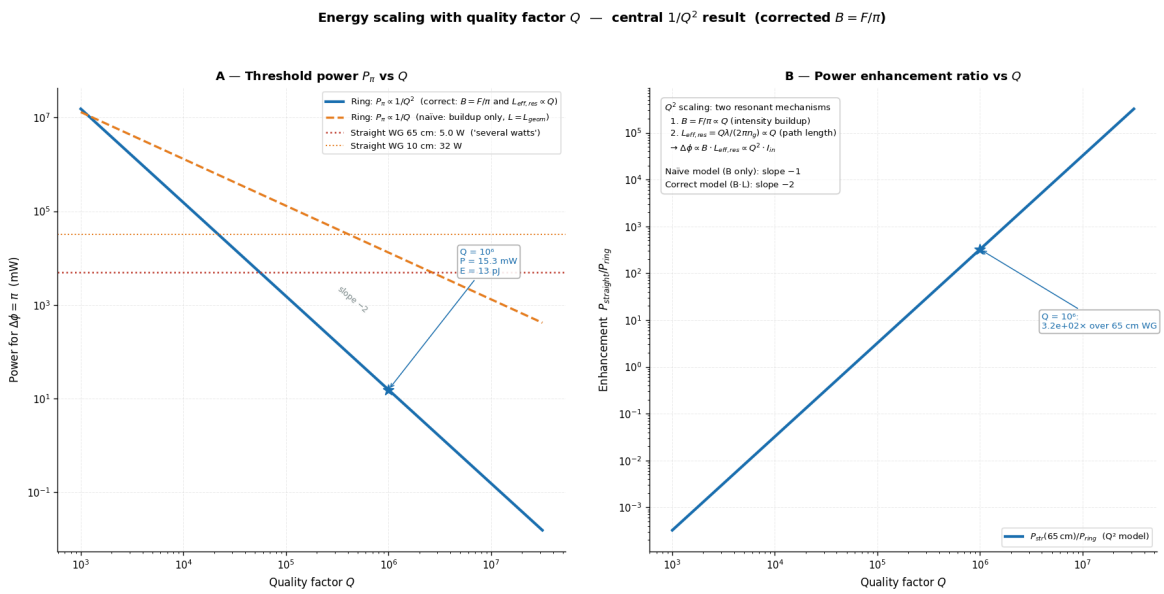


Figure 4

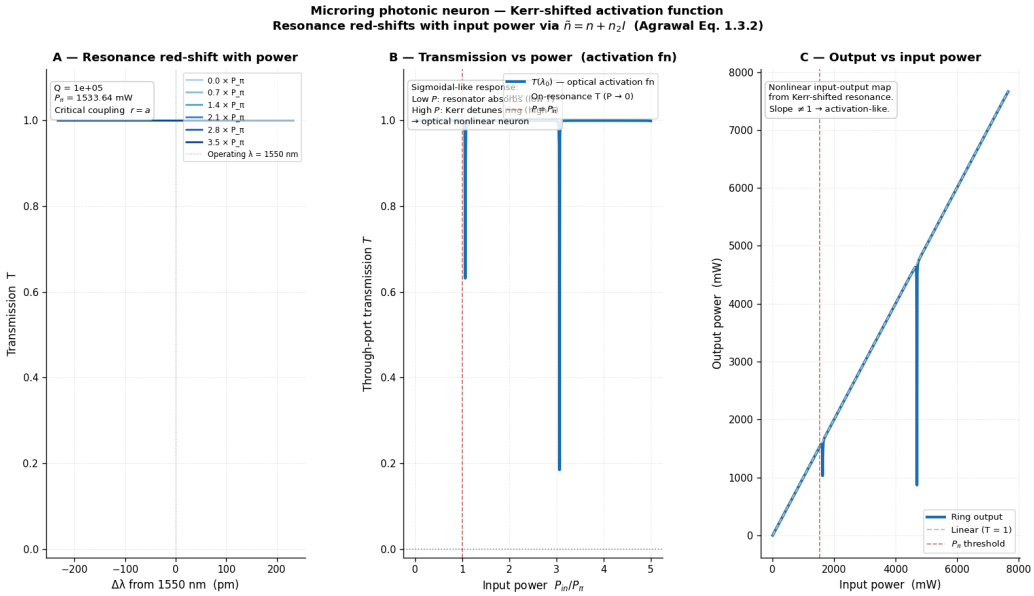


Figure 5

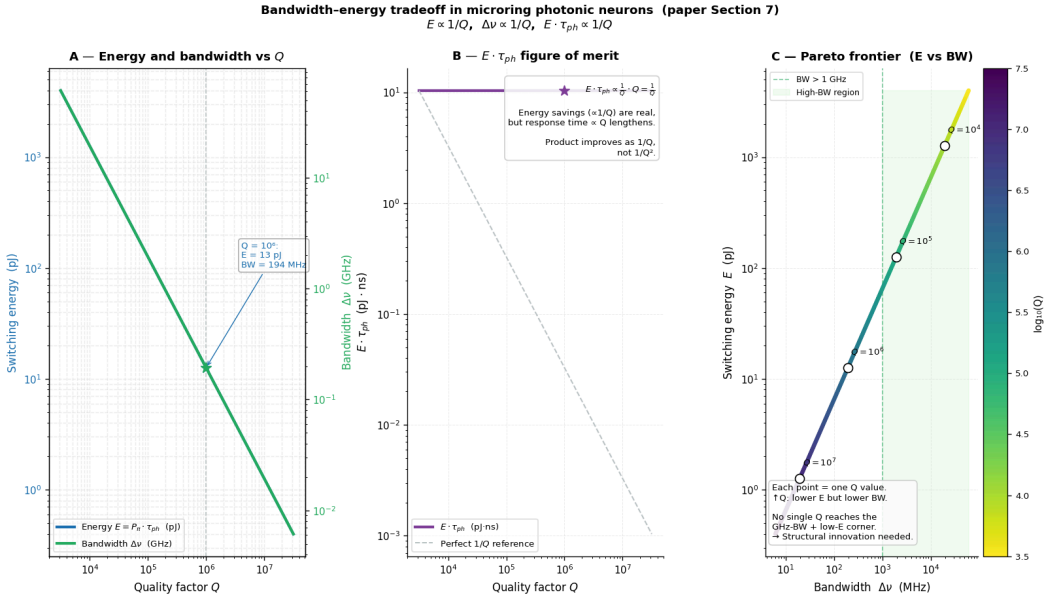


Figure 6

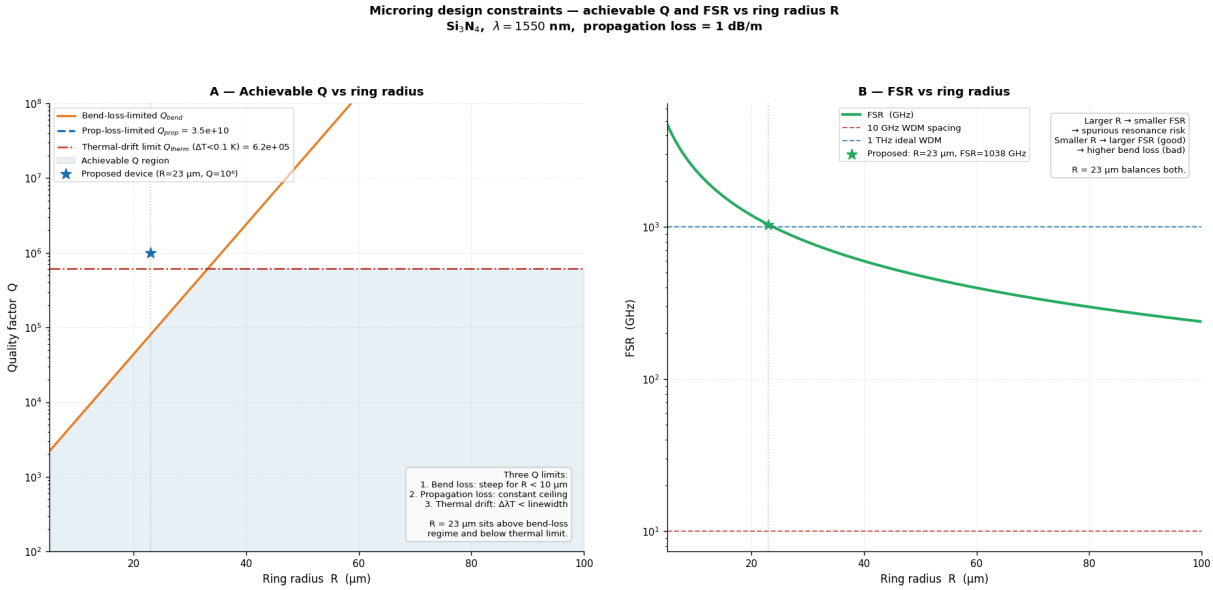
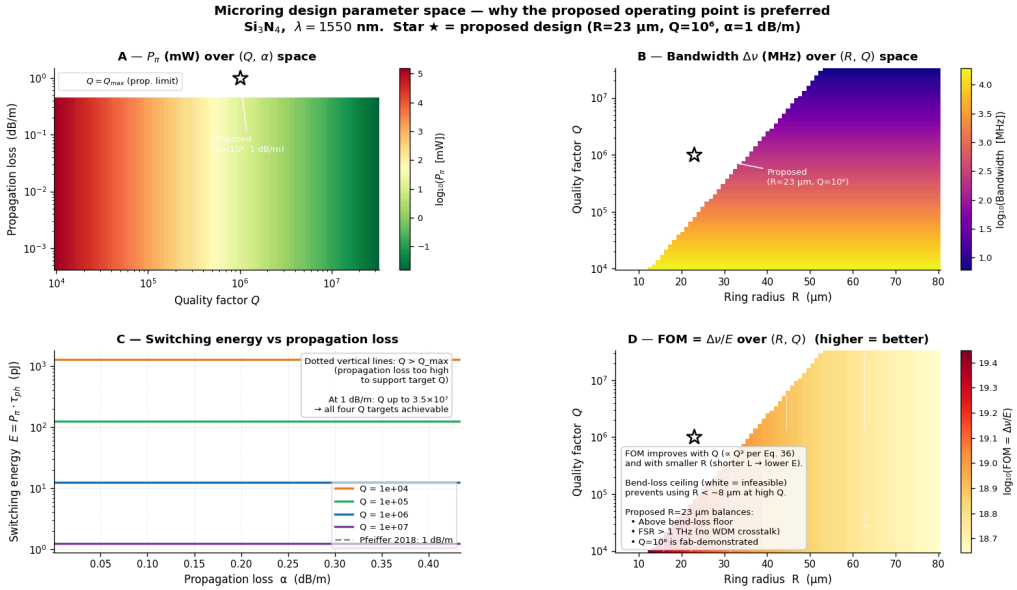


Figure 7


Microring design parameter space — why the proposed operating point is preferred
Si3N4, λ = 1550 nm. Star ★ = proposed design (R=23 μm, Q=10⁶, α=1 dB/m)
A — Pπ (mW) over (Q, α) space
Q = Qmax (prop. limit)
Proposed (Q=10⁶, 1 dB/m)
Propagation loss (dB/m)
Quality factor Q
log10(Pπ [mW])
B — Bandwidth Δν (MHz) over (R, Q) space
Proposed (R=23 μm, Q=10⁶)
Quality factor Q
Ring radius R (μm)
log10(Bandwidth [MHz])
C — Switching energy vs propagation loss
Dotted vertical lines: Q > Q_max (propagation loss too high to support target Q)
At 1 dB/m: Q up to 3.5×10⁷
→ all four Q targets achievable
Q = 1e+04
Q = 1e+05
Q = 1e+06
Q = 1e+07
Pfeiffer 2018: 1 dB/m
Switching energy E = Pπ · τph (pJ)
Propagation loss α (dB/m)
D — FOM = Δν/E over (R, Q) (higher = better)
FOM improves with Q (∝ Q² per Eq. 36) and with smaller R (shorter L → lower E).
Bend-loss ceiling (white = infeasible) prevents using R < ~8 μm at high Q.
Proposed R=23 μm balances:
• Above bend-loss floor
• FSR > 1 THz (no WDM crosstalk)
• Q=10⁶ is fab-demonstrated
Quality factor Q
Ring radius R (μm)
log10(FOM = Δν/E)